

SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO5: Design Pushdown Automata and analyze their equivalence with Context-Free Grammars through formal construction and proof techniques.
(BL : 3)

Assessment Tool Weightage: 10%

Total Marks:20

QUESTIONS: 20
Duration :1 Hour

Mock Interview question

Code of Conduct:

I, V. Smylika / 192342115 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

V. Smylika.
Signature of the student:

COS: ASSESSMENT TOOL 3

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What is a Pushdown Automaton? How is it different from a Finite Automaton?
A Pushdown Automaton is a computational model used to recognize context-free languages. It is similar to a finite Automaton but has an additional stack for storing information.

Difference:

Feature

Finite Automaton (FA)

Pushdown Automaton

Memory

No extra memory

Has a stack

Languages

Regular Languages

Context free languages

Power

Less powerful

More powerful

Example

a^*

$\{a^n b^n \mid n \geq 0\}$

Define the components of a PDA formally.

A PDA is formally represented by a 7-tuple.

$$PDA = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0, F\}$$

Where Q = finite set of states

Σ = input alphabet

Γ = stack alphabet

δ = Transition Function

q_0 = Initial state

Z_0 = Initial stack symbol

F = set of final states.

What is a stack in PDA? Why is it important?

A stack is a memory structure used by a PDA that follows the

The stack supports operations such as:

- push — add a symbol to the stack
- pop — remove the top symbol
- Read/Replace — inspect and replace the top symbol.

What is an Instantaneous Description in PDA?

An Instantaneous Description represents the current configuration of a particular movement.

It is represented as (q, w, y)

q = current state

w = remaining string

y = current stack

5. What are the different types of moves in a PDA?

The main types of PDA moves are:

1. Push move: A symbol is added to stack

3. Replace move: The top stack symbol is replaced.

2. Pop move: The top stack symbol is removed.

4. ϵ -move: The PDA changes state without consuming any input.

6. What is acceptance by final state and empty stack?

A PDA can accept a string in two common ways.

Acceptance by Final state: The input is completely consumed and PDA is in final state.

Acceptance by Empty stack: The input is completely consumed and stack is empty.

7. Define the Language Accepted by a PDA.

The Language Accepted by a PDA is the set of all strings that are accepted by the PDA according to its acceptance condition.

Acceptance by final state: $L(P) = \{ w \in \Sigma^* \mid (q_0, w, z_0) \vdash^* (q_f, \epsilon, y), q_f \in F \}$

8. How does a PDA process a string using IDs?

A PDA processes a string through a sequence of Instantaneous Descriptions (IDs).

For ex: The language $\{a^n b^n\}$. the PDA:

1. Reads each a .

2. Pushes an A onto the stack.

3. When b 's begin, it pops one A for every B .

4. Thus the string is accepted.

Difference between Deterministic PDA and Non-Deterministic PDA

Feature

Full form

choice of moves

Restrictions

DPDA

Deterministic PDA

At most one move

Restricted.

Context free contents

Why are PDAs equivalent to Context-Free Grammars?
PDAs and CFGs are equivalent in terms of the languages they can describe.
That means:

- Every CFG can be converted into a PDA
 - Every PDA can be converted into a CFG.
- Both recognize exactly the class of context-free languages (CFLs).
∴ $CFG \leftrightarrow PDA \leftrightarrow \text{Context-free Languages}$.

11. What types of Languages are accepted by PDA? Give examples.
A PDA accepts context-free languages (CFLs)

Example include:

1. $L = \{a^n b^n \mid n \geq 0\}$ 2. Balanced parentheses: $\{(, (,), (,), \dots\}$ 3. $L = \{a^n b^n c^n \mid n \geq 0\}$

PDAs are commonly used for languages involving nested or matching structures.

12. What is a Turing Machine? How is it more powerful than PDA?

A Turing Machine (TM) is a mathematical model of computation consisting of a read/write head, finite set of states, and transition rules.

- PDA has a stack, while a TM has an effectively unlimited read/write tape.
- A PDA can recognize context-free languages.

$$L = \{a^n b^n c^n \mid n \geq 0\}$$

13. Define the components of a Turing Machine.

A Turing Machine is formally defined as a 7-tuple $TM = \{Q, \Sigma, \Gamma, \delta, q_0, \text{Final state}, B\}$

Where:

- Q = Finite set of states
- Σ = Input Alphabet
- Γ = Tape Alphabet
- δ = Transition function
- q_0 = Initial state
- B = Blank symbol

14. Explain tape, head, and state transitions in a Turing Machine.

Tape: The tape is an unbounded memory divided into cells.

Head: The Head reads the symbol in the current cell and

16. what is storage in finite control in Turing Machines?
 Storage in finite control refers to the information that can be remembered the current state of Turing Machine.

For large or unbounded information, the TM uses its tape instead.

17. Compare FA, PDA, and TM based on memory, power and language acceptance.

Feature	FA	PDA	TM
Memory	Finite control	stack	unlimited tape
Read/Write	Read Input	Reads input + stack	Reads / writes tape
Power	Lowest	Medium	Highest
Languages	Regular	Context-free	Recursively Enumerable
Ex:	a^*	$\{a^n b^n\}$	$\{a^n b^n c^n\}$

18. what is DFA?

A DFA is a Finite Automaton in which, for every state and every input symbol there is exactly one Transition.

$DFA = \{Q, \Sigma, \delta, q_0, F\}$

Q = set of states

q_0 = Initial state

Σ = Input alphabet

F = Final states

δ = Transition function

19. which language is Accepted by a Turing Machine?

A Turing Machine can recognize Recursively Enumerable languages also called

Turing-Recognizable Languages.

20. How are Turing Machines Applied in real-world Applications?

Although a Turing Machine is mainly a theoretical model, its ideas form the basis of modern computing.

Applications include:

- Compiler Design
- Algorithm Development
- Computer Science
- Artificial Intelligence
- Data Processing
- Software Verification

ϵ -transitions

start with 1.

$x_1, x_3 = 10^{10}$

$y_1, y_2 = 10^1$

SIMATS ENGINEERING ASSESSMENT TOOLS

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100